

**CAFFEINE & CADAVER**

MBBS BIOCHEMISTRY NOTES SERIES

PYQ 2010–2025

MBBS BIOCHEMISTRY · MODULE 01

Enzyme Inhibition

Kinetics · Inhibitors · Clinical Applications

SUBJECT**Biochemistry — Paper 1****EXAM RELEVANCE****Core Curriculum Theory & Viva****PYQ FREQUENCY****Standard University Exam High-Yield**

★ APPEARED IN MBBS – YEARS

2010-S

2013

2013-S

2014-S

2015

2016

2017

2018

2019

2019-S

2021

2025

01 The Big Picture

🔗 WHY THIS MATTERS CLINICALLY

Enzyme inhibition is not just theory — it is the basis of **almost every drug in medicine**. Statins inhibit HMG-CoA reductase. Methotrexate inhibits DHFR. Aspirin irreversibly inhibits COX. Allopurinol is a suicide inhibitor of xanthine oxidase. Understanding inhibition kinetics tells you *how drugs work, why they fail, and how resistance develops*.

02 Enzyme Catalysis & Kinetics Derivation

🔑 THERMODYNAMIC PRINCIPLES

Enzymes are biological catalysts that accelerate chemical reactions by **stabilizing the transition state** and lowering the **activation energy ($\Delta G^\ddagger$)**. They do not alter the overall free energy change (ΔG) or shift the equilibrium position of the reaction; they simply allow the system to reach equilibrium much faster.

The catalytic event occurs within the **Active Site**, a highly specialized three-dimensional cleft or pocket lined with catalytic and binding amino acid residues. Active site binding is explained by Emil Fischer's classic *Lock-and-Key Model* (rigid complementarity) and Daniel Koshland's modern *Induced Fit Model* (dynamic conformational change upon ligand binding).

THE BRIGGS-HALDANE STEADY-STATE DERIVATION

To understand how enzyme velocity behaves mathematically, we derive the fundamental **Michaelis-Menten Equation** using the Briggs-Haldane steady-state approximation.

// THE REACTION SCHEME

 Enzyme + Substrate
(E + S)
 $\rightleftharpoons (k_1 / k_{-1})$
 Enzyme-Substrate Complex
(ES)
 $\rightarrow (k_2)$
 Enzyme + Product
(E + P)


The Steady-State Assumption: The concentration of the transition state [ES] remains constant over time. Therefore, the rate of ES formation equals the rate of ES breakdown:

$$k_1 [E] [S] = (k_{-1} + k_2) [ES]$$



Defining the Michaelis Constant (Km): By grouping the rate constants:

$$K_m = (k_{-1} + k_2) / k_1$$

Substitute Km back: $[E][S] = K_m[ES]$.



Conservation of Enzyme: Since free enzyme [E] cannot be measured directly, we express it in terms of total enzyme [E]_t:

$$[E] = [E]_t - [ES]$$

Substituting this into the Km equation yields: $([E]_t - [ES])[S] = K_m [ES]$.



Solving for [ES] & Velocity: Solving for [ES] gives: $[ES] = ([E]_t[S]) / (K_m + [S])$. Since velocity $v = k_2[ES]$ and maximum velocity $V_{max} = k_2[E]_t$, we multiply both sides by k_2 to arrive at the **Michaelis-Menten Equation**:

$$v = (V_{max} \cdot [S]) / (K_m + [S])$$

This hyperbolic equation explains that at low [S] ($[S] \ll K_m$), velocity is first-order, while at high [S] ($[S] \gg K_m$), the enzyme is saturated and velocity is zero-order ($v \approx V_{max}$).

LINEWEAVER-BURK LINEARIZATION

Taking the double reciprocal of the Michaelis-Menten equation yields the linear **Lineweaver-Burk Equation**:

$$1/v = (K_m/V_{max}) \cdot (1/[S]) + 1/V_{max}$$

This allows researchers to plot experimental data as a straight line ($y = mx + c$), where the Y-intercept represents $1/V_{max}$ and the X-intercept represents $-1/K_m$.

03 Allosteric Regulation & Cooperativity

MECHANICS OF ALLOSTERY

Not all enzymes obey classic Michaelis-Menten kinetics. **Allosteric enzymes** are oligomeric proteins (multiple subunits) that exhibit **cooperativity** – where the binding of a ligand to one active site influences the binding properties of other active sites.

Allosteric enzymes display a characteristic **sigmoidal (S-shaped) saturation curve** rather than a hyperbola. They bind effectors at a distinct regulatory pocket called the **allosteric site**, causing conformational shifts that either activate (positive cooperative) or inhibit (negative cooperative) the enzyme.

THE TWO MODELS OF COOPERATIVITY

Medical biochemistry defines two prominent models explaining allosteric shifts:

FEATURE	CONCERTED MODEL (MWC MODEL)	SEQUENTIAL MODEL (KNF MODEL)
PROPONENTS	Monod, Wyman, and Changeux	Koshland, Némethy, and Filmer
SUBUNIT STATES	All subunits must exist in the same state (either all Tense [T] or all Relaxed [R]). No hybrid states allowed.	Subunits can change states individually . Hybrid states (some T, some R) are common.
SYMMETRY RULE	Symmetry is preserved. Binding of a substrate shifts the equilibrium of the entire oligomer between the T-state (low affinity) and R-state (high affinity).	Symmetry is not preserved. Substrate binding to one subunit induces a conformational shift only in that subunit, which then sequentially alters neighbors.
EFFECTOR ACTION	Activators stabilize the R-state; inhibitors stabilize the T-state, shifting the S-curve left or right.	Binding increases or decreases affinity of neighboring active sites sequentially via induced fit.

04 Covalent Modification & Zymogen Activation

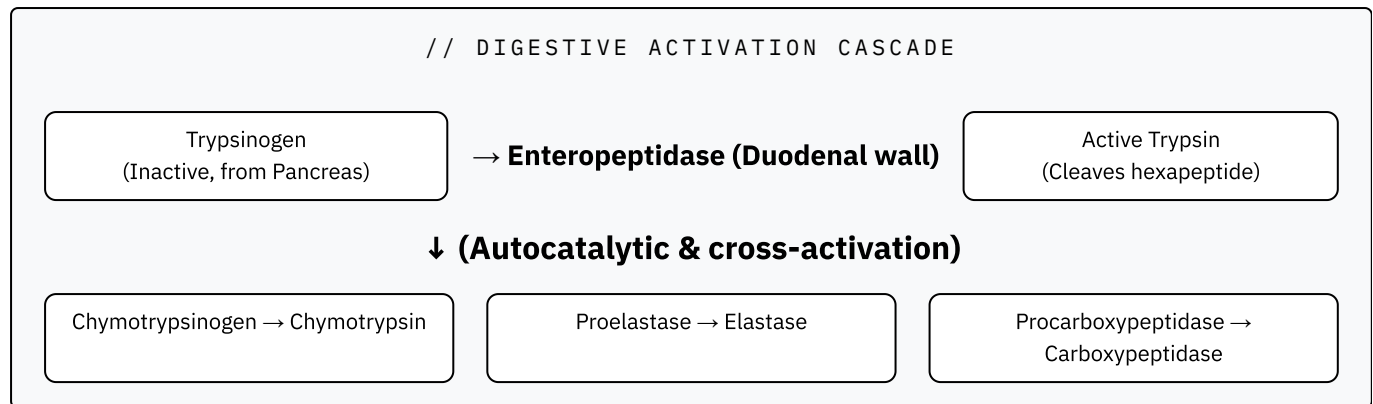
POST-TRANSLATIONAL REGULATION

In addition to allosteric effectors, the body uses two highly efficient molecular mechanisms to regulate enzyme activity on demand:

- **Reversible Covalent Modification:** The addition or removal of a chemical group (most commonly phosphate, via *kinases* and *phosphatases*) to specific serine, threonine, or tyrosine residues of the enzyme. For example, glycogen phosphorylase is active when phosphorylated, while glycogen synthase is inactive when phosphorylated.
- **Zymogen Activation (Proteolytic Cleavage):** The irreversible activation of an inactive enzyme precursor (zymogen/proenzyme) by the selective cleavage of one or more peptide bonds. This is crucial for digestive enzymes and blood coagulation factors, preventing them from destroying the cells that synthesize them.

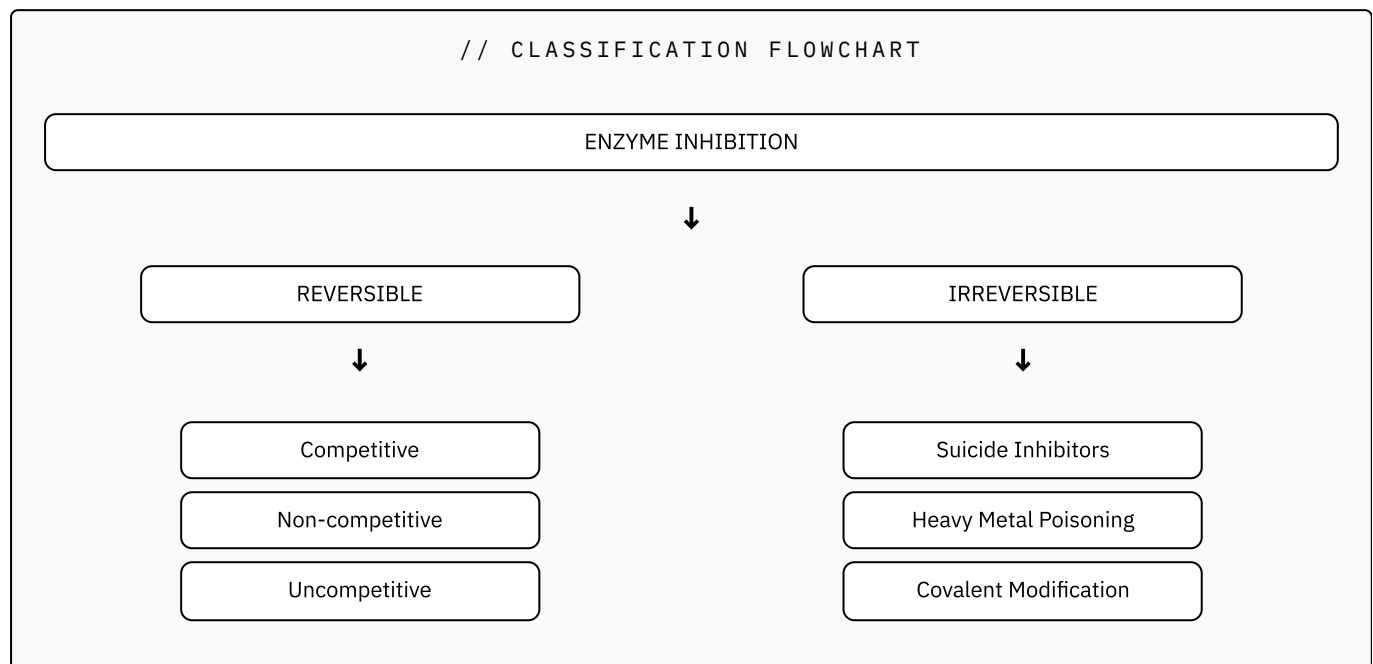
ZYMOGEN CASCADES IN DIGESTION

The classic medical example of zymogen activation is the activation cascade in the duodenum:



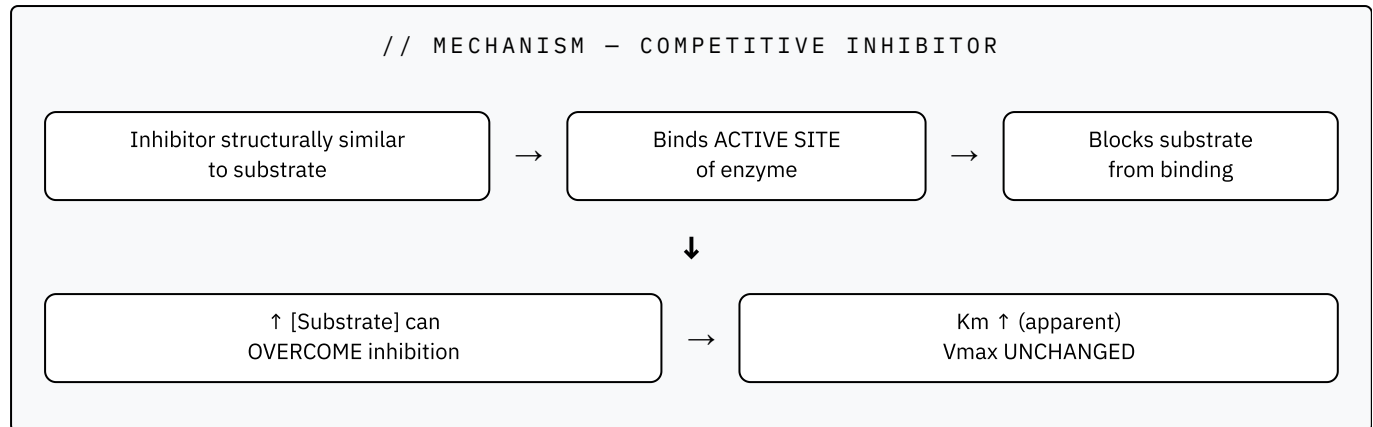
Viva Pearl: If trypsin is prematurely activated inside the pancreas, it causes a catastrophic self-digestion of the organ, clinically presenting as **acute hemorrhagic pancreatitis**. The pancreas protects itself by synthesizing *Pancreatic Trypsin Inhibitor (PSTI)* to block any trace trypsin.

05 Types of Inhibition



06 Competitive Inhibition ★ EXAMINER FAV

MECHANISM



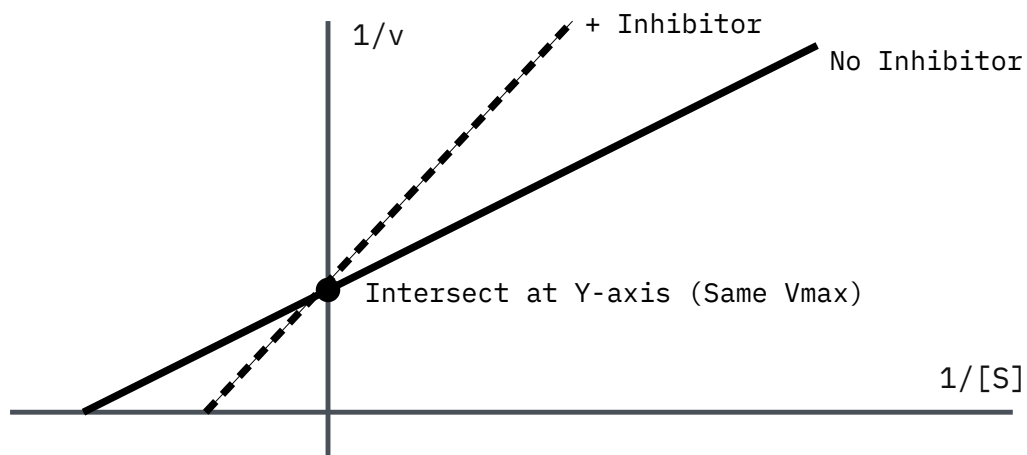
The Golden Rule: In competitive inhibition — **K_m increases** (more substrate needed to achieve half-V_{max}) but **V_{max} remains the same** because high substrate concentration can displace the inhibitor.



Lineweaver-Burk Plot: Lines **intersect on the Y-axis** (same V_{max}). X-intercept shifts toward origin → K_m increased. The two lines must cross the Y-axis at the same point — if they don't, it's NOT competitive.

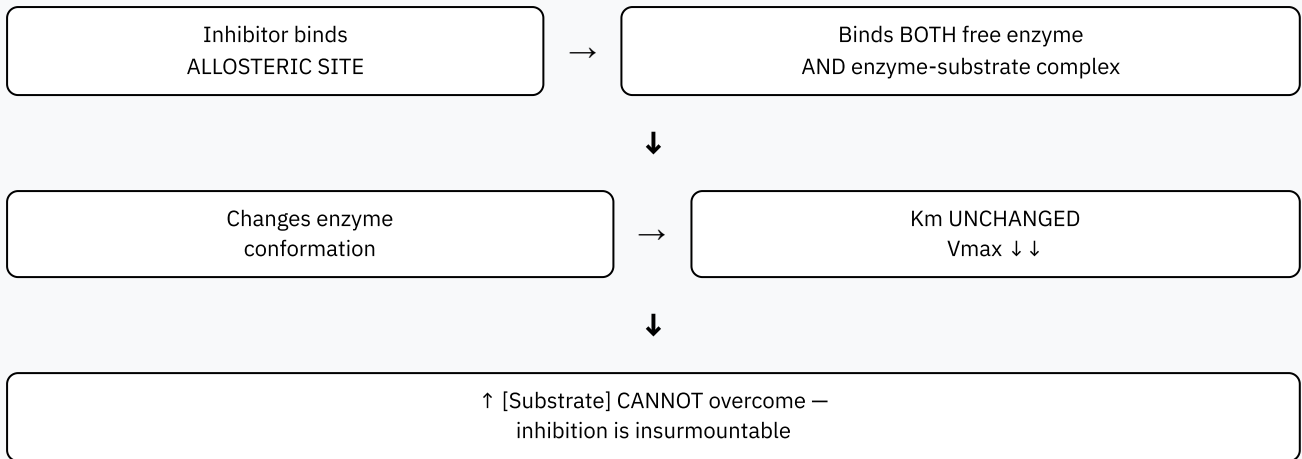


LINEWEAVER-BURK PLOT: COMPETITIVE



07 Non-Competitive Inhibition

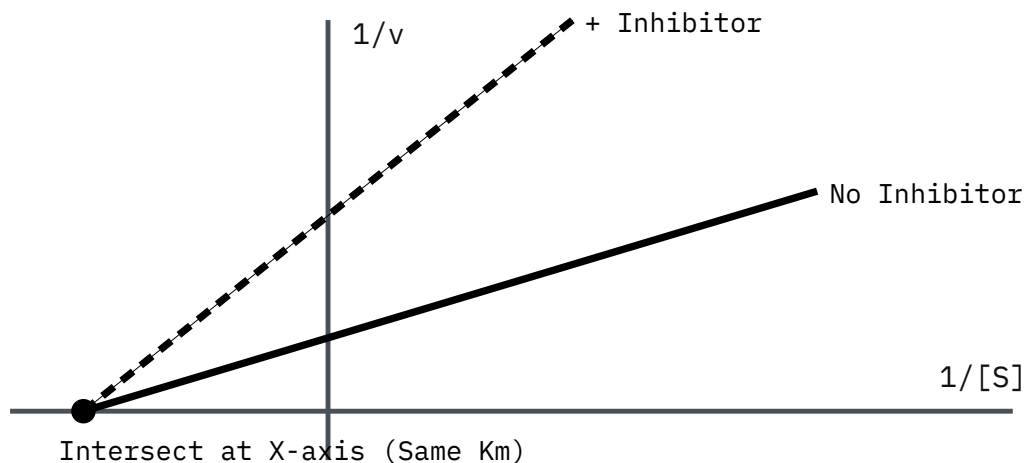
// MECHANISM – NON-COMPETITIVE INHIBITOR



Lineweaver-Burk Plot: Lines **intersect on the X-axis** (same K_m / same $-1/K_m$). Y-intercept increases $\rightarrow$ V_{max} decreased. Classic: parallel shift? No — they converge at X-axis.



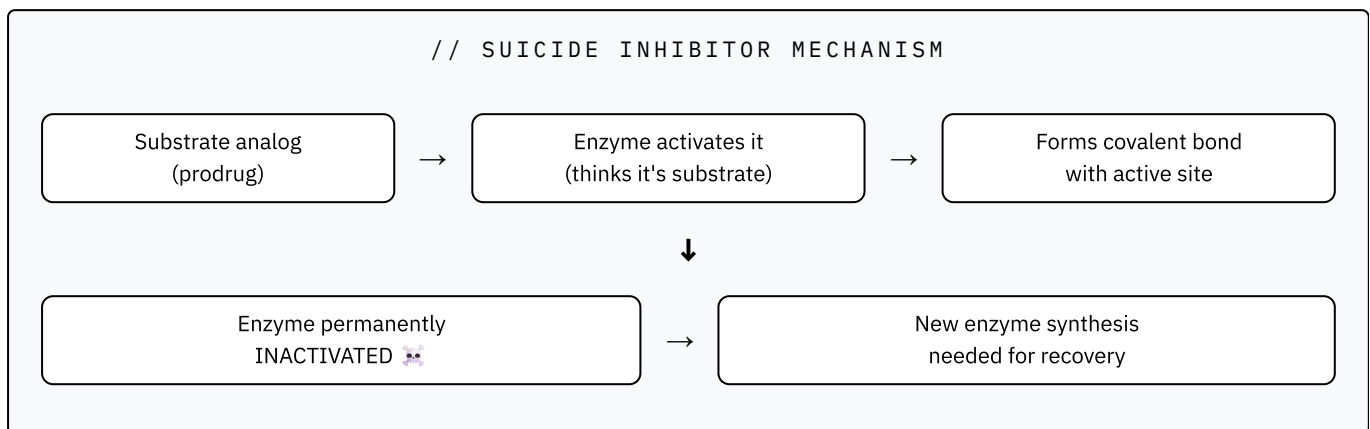
LINEWEAVER-BURK PLOT: NON-COMPETITIVE



08 Master Comparison Table ★ EXAMINER FAV

PARAMETER	COMPETITIVE	NON-COMPETITIVE	UNCOMPETITIVE
BINDING SITE	Active site	Allosteric site (both E and ES)	Only ES complex
EFFECT ON K_M	↑ K _m (apparent)	No change	↓ K _m (apparent)
EFFECT ON V_{MAX}	No change	↓ V _{max}	↓ V _{max}
OVERCOME BY ↑ [S]?	YES ✓	NO X	NO X
L-B PLOT INTERCEPT	Y-axis (same V _{max})	X-axis (same K _m)	Parallel lines (both shift)
K_I EXPRESSION	$K_m(\text{app}) = K_m(1 + [I]/K_i)$	$V_{\text{max}}(\text{app}) = V_{\text{max}}/(1 + [I]/K_i)$	Both K _m and V _{max} ↓ equally
CLASSIC EXAMPLE	Methotrexate (DHFR) Statins (HMG-CoA red.) Sulfanilamide (PABA)	Heavy metals (Hg, Pb, As) — specifically bind sulfhydryl (-SH) groups Cyanide (Cyt oxidase)	Lithium (inositol monophosphatase)

09 Irreversible & Suicide Inhibitors



DRUG	ENZYME INHIBITED	CLINICAL USE
ALLOPURINOL	Xanthine oxidase	Gout treatment
ASPIRIN (IRREVERSIBLE)	COX-1 / COX-2	Anti-platelet, anti-inflammatory
PENICILLIN	Transpeptidase (DD-carboxypeptidase)	Antibiotic — cell wall synthesis
EFLORNITHINE	Ornithine decarboxylase	Antiprotozoal (Trypanosomiasis)
5-FLUOROURACIL	Thymidylate synthase	Cancer chemotherapy

10 Km & Vmax — The Core Concept ★ EXAMINER FAV

MICHAELIS-MENTEN EQUATION

$$v = (V_{\max} [S]) / (K_m + [S])$$

Where: **v** = reaction velocity, **V_{max}** = maximum velocity, **[S]** = substrate concentration, **K_m** = Michaelis constant



K_m Definition: Substrate concentration at which reaction velocity = $\frac{1}{2}$ **V_{max}**. It reflects the *affinity* of enzyme for substrate. **Low K_m = High affinity** (less substrate needed). **High K_m = Low affinity**.



Hexokinase vs Glucokinase (Classic PYQ!): Hexokinase has **low K_m** for glucose — works even at low blood glucose (brain, RBC). Glucokinase has **high K_m** — only active after a meal when glucose is high (liver, pancreatic β cells). This is why glucokinase acts as a "glucose sensor."

11 Isoenzymes — LDH & CK ★ EXAMINER FAV

// LDH ISOENZYMES — TISSUE DISTRIBUTION

LDH-1
(H₄)

Heart
RBCs
↑ in MI

LDH-2
(H₃M)

Heart
Kidney
↑ in MI

LDH-3
(H₂M₂)

Lungs
Spleen
Pulmonary

LDH-4
(H₁M₃)

Kidney
Placenta

LDH-5
(M₄)

Liver
Skeletal muscle
↑ in liver disease



LDH Flipped Pattern in MI: Normally, the concentration of **LDH-2 > LDH-1**. However, during a Myocardial Infarction (MI), massive release from damaged heart tissue causes the ratio to invert, resulting in a "flipped pattern" where **LDH-1 > LDH-2**.

// CK ISOENZYMES

CK-BB
(CK-1)

Brain

CK-MB
(CK-2)

Heart ← **MI marker**

CK-MM
(CK-3)

Skeletal muscle



CK-MB > 6% of total CK = highly suggestive of Myocardial Infarction. Rises within 4–6 hours, peaks at 12–24 hours. Separated by electrophoresis or immunoassay.

12 Clinical Correlations

Drug–Enzyme Inhibition Pairs (High Yield)

- **Methotrexate** → Competitive inhibitor of Dihydrofolate Reductase (DHFR) → Used in cancer, rheumatoid arthritis, psoriasis
- **Statins** (Atorvastatin, Rosuvastatin) → Competitive inhibitor of HMG-CoA Reductase

RATE LIMITING STEP

 of cholesterol synthesis → Hypolipidemia treatment
- **Allopurinol** → Suicide inhibitor of Xanthine Oxidase → Reduces uric acid → Gout treatment
- **Aspirin** → Irreversible inhibitor of COX-1 & COX-2 via acetylation of a specific serine residue at the active site → Anti-platelet effect lasts life of platelet (7–10 days)
- **Sulfanilamide / Sulfonamides** → Competitive analog of PABA → Inhibits bacterial dihydropteroate synthase → Antibiotic
- **Penicillin** → Irreversible inhibition of Transpeptidase → Blocks bacterial cell wall synthesis
- **Neostigmine** → Reversible inhibitor of Acetylcholinesterase → Used in myasthenia gravis
- **Organophosphates** (insecticides) → Irreversible inhibition of AChE → Toxicity: SLUDGE syndrome

13 Memory Mnemonics

MNEMONIC 1

"Competitive Kills Km, Non-competitive Nails Vmax"

Competitive → Km changes ($\uparrow$), Vmax stays.

Non-competitive → Vmax changes ($\downarrow$), Km stays.

MNEMONIC 2 – LINEWEAVER-BURK

"Competitive meets at Y, Non-competitive meets at X"

Competitive inhibition → lines cross at Y-axis intercept (same $1/V_{max}$ = same Vmax).

Non-competitive → lines cross at X-axis intercept (same $-1/K_m$ = same Km).

MNEMONIC 3 – CK ISOENZYMES

"Brain Beats Muscles" → BB, MB, MM

CK-BB = Brain | CK-MB = Myocardium (Heart) | CK-MM = Muscles

CK-MB is the GOLD STANDARD cardiac marker in early MI (before Troponin era in many exams).

 MNEMONIC 4 – SUICIDE INHIBITORS

"APPLE" – Always Permanently Prohibit Liver Enzymes

Allopurinol (xanthine oxidase) · Penicillin (transpeptidase) · 5-FU (thymidylate synthase) · L-DOPA analog · Eflornithine (ODC)

 VIVA RAPID FIRE – ENZYME INHIBITION

Q1 What happens to K_m and V_{max} in competitive inhibition?

K_m increases (apparent), V_{max} remains unchanged. The inhibitor competes with substrate for the active site. High substrate concentration can displace the inhibitor.

Q2 Why is allopurinol called a suicide inhibitor?

Allopurinol is first oxidized by xanthine oxidase to alloxanthine (oxypurinol), which then permanently and covalently binds to the active site of xanthine oxidase, inactivating it irreversibly. The enzyme participates in its own inhibition – hence "suicide."

Q3 Why is K_m of glucokinase higher than hexokinase?

Glucokinase (hepatic) is a low-affinity, high- K_m enzyme that functions as a glucose sensor – it activates only when blood glucose is high (postprandial). Hexokinase has high affinity (low K_m) and works even at basal glucose levels, ensuring glucose supply to brain and RBCs at all times.

Q4 What does the Lineweaver-Burk plot show?

It's a double reciprocal plot of $1/v$ vs $1/[S]$. Y-intercept = $1/V_{max}$. X-intercept = $-1/K_m$. Slope = K_m/V_{max} . Used to distinguish types of inhibition: competitive lines cross at Y-axis; non-competitive lines cross at X-axis.

Q5 Name four non-functional plasma enzymes and their significance.

1) ALT/SGPT – liver damage (hepatocellular). 2) AST/SGOT – liver, cardiac damage. 3) Alkaline Phosphatase – obstructive jaundice, bone disease. 4) Amylase – pancreatitis. 5) CK-MB – Myocardial infarction. These enzymes normally function intracellularly; their appearance in plasma indicates cell damage.

Q6 Differentiate inhibitors from uncouplers (ETC context – frequent combo Q)

Inhibitors block electron transport at specific complexes (e.g., CN^- at Complex IV, Rotenone at Complex I) – stop both electron flow AND ATP synthesis. Uncouplers (e.g., 2,4-DNP, thermogenin) allow electron transport to continue but dissipate the proton gradient – electrons flow but NO ATP is made. Heat is generated instead.

15 Group-D Essay Questions – How to Answer

 EQ FORMULA (YOUR TEACHER'S FLOWCHART TRICK)

For every Group-D (Essay/Justify) answer, use this structure:

// EQ ANSWER TEMPLATE – "METHOTREXATE IS A COMPETITIVE ENZYME INHIBITOR"

Define

What is competitive inhibition?



Mechanism Flowchart

MTX → mimics DHFA → binds DHFR active site → blocks tetrahydrofolate synthesis



Kinetics

K_m ↑, V_{max} unchanged
Overcome by ↑ substrate



Clinical Consequence

↓ THF → ↓ purine synthesis → ↓ DNA synthesis → Anti-cancer / Anti-rheumatic



Rescue

Leucovorin (folinic acid) bypasses the block – used to rescue normal cells

MBBS Biochemistry – Module 01: Enzyme Inhibition

Compiled from PYQ 2010–2025

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